MILITARY SPECIFICATION

THREAD, PARA-ARAMID, SPUN, INTERMEDIATE MODULUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for spun para-aramid, intermediate modulus thread used for machine and hand sewing.

1.2 Classification.

1.2.1 Sizes. Sizes shall be as specified (see 6.2) and in accordance with table I.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-1183	-	Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS), Rigid
PPP-P-50	-	Packaging and Packing of Thread for Domestic and Overseas Shipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8310

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Shades for Sewing Threads 1968

(Application for copies should be addressed to the Color Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2146 - Propylene Plastic Molding and Extrusion Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Government and contractor purchases. The requirements specified in 3.6, 3.8, 3.9, and the related provisions of section 4, and all of section 5 apply only to thread purchased directly by the Government. All other requirements apply both to thread purchased by a contractor as a component for an end item and to thread purchased directly by the Government.

3.2 Standard sample. When a standard sample is established (see 6.3), the thread shall match the standard for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced.

3.3 Material.

3.3.1 Para-aramid cut staple. The cut staple fibers shall be made from para-aramid filament having a linear density of 1.5 denier per filament. The cut staple fibers shall have a nominal length of 1.5 inches. The para-aramid fibers shall not carbonize at a temperature below 675°F.

3.3.2 Yarn. The yarn for the individual plies of the thread shall be spun from the cut staple fibers specified in 3.3.1 (see 4.2.2).

3.4 Color. The color (shade) shall be as specified in the applicable end-item document or in the contract or purchase order (see 4.2.2 and 6.2). Shade designations by letter (e.g. - Olive Drab S-1), and related cable numbers (e.g. C.A.66022), refer to the DOD Standard Shade for Sewing Threads 1968 (see 2.2).

3.4.1 Visual color matching. The shade of the end item thread, after removal of finish when applicable, shall match the specified applicable shade standard (see 6.3), when tested as specified in 4.2.4.

3.4.2 Colorfastness. The dyed and finished thread shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the dyed and finished thread shall show "good" fastness to laundering (after 3 cycles). Testing shall be as specified in 4.2.4.

3.5 Physical requirements. The finished thread shall conform to the applicable requirements for the specified size set forth in table I when tested as specified in 4.2.4.

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TABLE I. Physical requirements

Size	Plies	Final twist Minimum (Turns per inch)	Length per pound (Yards) Min	Length per pound (Yards) Max	Approx-imate Tex	Breaking strength Minimum Pounds	Breaking strength Minimum Newton(N)	Elongation Maximum (percent)
T-35	2	11.60	11,970	13,230	41	6.0	26.7	6
T-50	3	9.50	7,980	8,820	59	8.5	37.8	6
T-70	4	8.20	5,700	6,300	82	12.0	53.4	6
T-90	5	7.35	4,560	5,040	98	16.0	71.2	6
T-105	6	6.75	3,990	4,410	118	19.0	84.5	6
T-135	7	6.20	3,420	3,780	138	22.0	97.9	6

3.5.1 Twist (turns per inch). Each of the individual plies shall be twisted initially with not less than the number of turns per inch to be used in the final twist. The final plied twist shall not be less than the applicable minimum turns per inch specified in table I.

3.5.2 Twist direction. Unless otherwise specified (see 6.2), the direction of the twist for single ply shall be "S" and for the plied thread shall be "Z", when tested as specified in 4.2.4.

3.5.3 Aging resistance. Each size of the finished thread shall retain a minimum of 85 percent of its original breaking strength when subjected to oven aging at 300 ± 10°F for 4 hours when tested as specified in 4.2.4.

3.6 Knots. The thread shall average not more than one full thread knot per 8 ounces.

3.7 Finish. The thread shall have a soft finish and shall contain only the minimum amount of non flame propogating lubricant to facilitate sewing (see 4.2.2).

3.8 Put-up. Unless otherwise specified (see 6.2), the thread shall be put-up as specified in table II on commercial holders (see 3.8.1) on a weight per holder basis. The average weight per holder shall not be less than the specified nominal weight minus 3.0 percent tolerance when examined (on a 10-holder basis) as specified in 4.2.3.3. The thread shall be put-up in one continuous length per holder and shall be so wound that each turn and layer is free from entanglement when examined as specified in 4.2.3.2.

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TABLE II. Put-up on holders

Thread size	Type of holder	Traverse	Nominal weight (ounces)
T-35	5 inch cone or tube	4-inch	8
	7 inch cone or tube	6-inch	24
T-50	5 inch cone or tube	4-inch	8
	5 inch cone or tube	4-inch	12
	7 inch cone or tube	6-inch	24
T-70	5 inch cone or tube	4-inch	8
	7 inch cone or tube	6-inch	24
T-90	7 inch cone or tube	6-inch	12
	7 inch cone or tube	6-inch	24
T-105	7 inch cone or tube	6-inch	24
T-135	7 inch cone or tube	6-inch	24

3.8.1 Plastic tube holders. When plastic tube holders are used they shall be approved by the contracting officer. They shall be single-head plastic tubes made of either acrylonitrile or polypropylene which shall conform to the following requirements:

3.8.1.1 Acrylonitrile. The tubes shall be acrylonitrile-butadiene-styrene (ABS) rigid plastic (see 6.4) meeting the requirements for type I of L-P-1183 with the following values substituted in the table of property values.

Property values

Impact strength (Izod), ft-lb per inch of notch, min.	
At 23° ± 2°C (1/8 inch sample)	1.5
Tensile yield stress, p.s.i., min.	6,000
Deflection temperature under load (264 p.s.i. fiber stress), min.	190°F (88°C)
Modulus of elasticity in tension, p.s.i., min.	390,000
Rockwell hardness (R scale), min.	105
Specific gravity, 23°/23°C, max.	1.2
Chemical resistance 40 hours, immersion in heptane at 23° ± 2°C, weight change, percent, max.	5.0

3.8.1.2 Polypropylene. The tubes shall be molded from a virgin propylene homopolymer plastics (see 6.5) meeting the requirements of classification I-58208 under ASTM D 2146.

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3.8.2 Ready-made lockstitch bobbins. When thread is put up on lockstitch bobbins, the style of bobbin and length of thread thereon shall be as specified in the contract or purchase order (see 6.2). Unless otherwise specified (see 6.2), ready-made lockstitch bobbins shall be put up in either one-half or one gross pack.

3.9 Identification marking. Except when commercial identification markings are specified in the invitation to bid, each holder shall have a label attached in such a manner as to remain in place and be clearly legible until all thread has been removed. The label shall be printed with the information as specified below.

National stock number
Weight (net)
Direction of twist
Color
Ply
Number size
Contract number
Date manufactured (month and year)
Name of contractor
Nomenclature

3.10 Workmanship. The finished thread shall conform to the quality established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.2.2 Certification. The contractor shall furnish a certificate of compliance with each shipment or lot stating the following:

a. The yarn for the individual plies of thread are spun from the cut staple fibers specified in 3.3.1 (see 3.3.2).

b. The color conforms to that specified in the contract or purchase order (see 3.4).

c. The thread has a soft finish and contains the minimum amount of non flame propogating lubricant to facilitate sewing (see 3.7).

d. When thread is put-up on single head plastic tubes, the tubes are made up of the material specified in either 3.8.1.1 or 3.8.1.2.

4.2.3 End item examination.

4.2.3.1 Thread on holder examination. Thread on the holder shall be examined for the defects listed in table III. The lot size shall be expressed in units of one holder. The sample unit shall be one holder. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE III. Thread on holder defects

Examine	Defect
Identification label	Missing, incorrect, incomplete, illegible, or insecurely attached.
Type of holder	Other than specified.
Surface condition	Loose or unattached ply. Cut, tear, chafe, slip, affecting strength of thread or interfering with easy location of end and initial unwinding.
Cleanliness	Dirt, spot, or stain, clearly noticeable.
Finish	Other than specified. Uneven, lumpy, bare or thin spots.

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4.2.3.2 Thread as unwound examination. The thread shall be examined while unwinding, for the defects listed in table IV. All defects shall be counted regardless of their proximity to each other. The lot size for holders shall be expressed in units of one holder and the sample unit shall be one holder. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE IV. Thread as unwound defects

Examine	Defect
Continuous length	Not in one continuous length.
Knots	Thread on a holder averages more than one thread knot per eight ounces (see 3.6).
Winding	Improperly or not firmly wound, resulting in kinks, knots, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of the thread.
Color	Uneven, apparent on successive layers or end to end.
Tackiness or adhesion	Strands adhere to each other or to holder, affecting unwinding tension.

4.2.3.3 Examination for net weight per holder. The sample for this examination shall be 10 wound holders, taken at random from the lot. The holders shall be weighed individually, and the net weight of thread determined. The values obtained shall be averaged, and this average shall represent the net weight per holder in the lot. The lot shall be unacceptable if the net weight per holder is less than the specified weight minus the 3.0 percent tolerance (see 3.8).

4.2.3.4 Length per ready-wound bobbin examination. Ten bobbins taken at random from the lot shall be measured individually. Measurement shall be made on a skein reel with an accurate measuring clock under normal winding tensions. The average of the values obtained shall be the length per bobbin in the lot. The lot shall be unacceptable if the length per bobbin is less than the amount specified minus the 3.0 percent tolerance (see 3.8).

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4.2.4 End item testing. The end item shall be tested as indicated in table V. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. The lot size shall be expressed in units of one holder. The sample unit shall be one wound holder or sufficient holders to provide enough thread for the applicable tests. The inspection level shall be S-1, and the AQL for each characteristic, expressed in terms of test failures per hundred units, shall be 6.5. The requirements specified in section 3 are applicable to the results of the determinations made on a sample unit for all characteristics. All test reports shall contain the individual values used in expressing the final result.

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method
Shade match	3.4.1	4.3.1
Colorfastness:		
Laundering (after 3 cycles)	3.4.2	5614 1/ 2/
Number of plies	3.5	Visual 3/
Length per pound	3.5	4010
Breaking strength (initial)	3.5	4100 4/
Elongation	3.5	4100 5/
Twist t.p.i.:		
Individual	3.5.1	4052 6/
Final	3.5, 3.5.1	4054 6/
Direction of twist	3.5.2	4050
Aging resistance	3.5.3	4.3.2

1/ The specimens must be dried after each of the 3 laundering cycles.

2/ The color transfer cloth evaluation shall not apply.

3/ One determination per sample unit shall be made and the result reported as "pass" or "fail".

4/ Except that five determinations shall be made per sample unit.

5/ To be determined simultaneously with breaking strength.

6/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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4.2.5 Packaging inspection. An examination shall be made in accordance with the provisions of PPP-P-50 to determine that packaging, packing and marking comply with the section 5 requirements.

4.3 Methods of inspection.

4.3.1 Shade evaluation.

4.3.1.1 Finish removal procedure. Before evaluation for shade matching, the sample of thread shall be wet drycleaned in accordance with Method 5622 of FED-STD-191 (disregarding reference to the standard sample). Excess solvent shall be removed by centrifuging or wringing. The sample shall then be rinsed in distilled water at 120° to 160°F and dried at a temperature not exceeding 180°F. The dried sample shall then be conditioned for a minimum of 4 hours prior to evaluation for shade match (see 6.3.1).

4.3.1.2 Shade matching procedure. The thread sample shall be wound or prepared for examination in a manner similar to that for the standard shade sample. The sample shall be compared to the standard sample under artificial daylight at 7000 ± 500 K, and less than a good match shall be a test failure. The sample shall also be compared to the standard sample under incandescent lamplight at 2850 ± 100 K, and less than a good approximation shall be a test failure.

4.3.2 Aging resistance test. Five specimens shall be used for the aging test. The specimens shall be exposed for 4 hours to a temperature of 500 ± 10°F using a circulating air oven. Upon removal, the specimens shall be conditioned at standard atmospheric conditions for 4 hours and then tested for breaking strength as specified in Method 4100 of FED-STD-191. The aging resistance shall be calculated as follows:

$$\text{Aging resistance (percent)} = \frac{\text{Average breaking strength after aging}}{\text{Initial breaking strength}} \times 100$$

The results shall be reported to the nearest 1 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Levels A, B and C. The thread shall be packaged in accordance with the applicable requirements of PPP-P-50.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

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5.2.1 Levels A, B and C. The thread shall be packed in accordance with the applicable requirements of PPP-P-50.

5.3 Marking. In addition to any special markings specified in the contract or purchase order, shipments shall be marked in accordance with applicable provisions of PPP-P-50.

6. NOTES

6.1 Intended use. The thread is intended for use in sewing aramid materials for construction of protective combat clothing, and other flight safety equipment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Size required (see 1.2.1).
(c) Color (shade) required (see 3.4).
(d) Direction of twist, if other than specified (see 3.5.2).
(e) Put-up required (see 3.8).
(f) Style of bobbin and length of thread for ready-made lockstitch bobbins (see 3.8.2).
(g) When ready-made lockstitch bobbins are to be put-up other than specified (see 3.8.2).
(h) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For Military shades, the standards for shade reference are those contained in the Standard Color Card of the Official Standardized Shades for Sewing Threads 1968 (see 2.2). For other shades, address the contracting activity issuing the invitation for bids or request for proposal. Standard samples frequently used by the Military (directly or for components of end items) have been made available to normal contractors as basic reference for the establishment of the contractor's secondary standards conforming to the characteristics of the official standards. The basic shade standards are those appearing in the DOD Color Card. Prime thread manufacturers who inadvertently have not been furnished standards for establishment of their colorfastness standards in accordance with this document may obtain samples for this purpose from the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, ATTN: DRDNA-VTC.

6.3.1 Shade matching evaluation. In cases where it is not considered feasible to remove the finish prior to shade matching, the thread may be evaluated on an "as is" basis upon approval of the contracting officer.

6.4 Material for acrylonitrile tubes. "ABS" material known as "Kralastic MM" has been used satisfactorily to make injection-molded tubes to meet the requirements specified in 3.8.1.1. Kralastic MM is made by the Naugatuck Chemical Division of the U.S. Rubber Company, Naugatuck, CT 06770.

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6.5 Material for polypropylene tubes. "EL Rexene PP 11-S-5" has been used satisfactorily to make injection-molded tubes meeting the requirements specified in 3.8.1.2. Resin is made by Dart Industries, Inc., P.O. Box 37, West 115 Century Road, Paramus, NJ 07652.

6.6 Source of supply for spun para-aramid thread. Thread made of Kevlar 29 para-aramid, staple 970, manufactured by the E. I. Du Pont de Nemours, & Co., Inc., can be obtained from the following sources:

American & Efird Mills, Inc., Mt. Holly, NC 28120
Atlantic Thread and Supply Co. Inc., Baltimore, MD 21238
Crown Threads of Carolina, Lincolnton, NC 28092
Synthetic Thread Mills, Bethlehem, PA 18018
The Robinson Thread Co. Inc., Worcester, MA 01613

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

Project No. 8310-0134

Review activities:

Army - MD
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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-T-44100 Thread, Para-Aramid, Spun, Intermediate Modulus

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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